

Exercise Sheet for Week 4 - Simply-typed λ -calculus

Assigned: Monday 21 October
Suggested Deadline: Monday 28 October

Questions marked with * are optional, more difficult challenges.

Question 1: Typing closed Church-style terms

For each of the following closed Church-style λ -terms, find its unique typing derivation or prove that it is untypable.

(a) $\lambda x : \mathbb{B}. x$

$$\frac{\overline{x : \mathbb{B} \vdash x : \mathbb{B}} \text{VAR}}{\vdash \lambda x : \mathbb{B}. x \text{ ABS}}$$

(b) **true**

$$\overline{\vdash \text{true} : \mathbb{B}} \text{ T}$$

(c) $\lambda x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}. \lambda y : \mathbb{B}. x y y$

$$\frac{\frac{\frac{\frac{\Pi_1 \quad \overline{x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash y : \mathbb{B}} \text{VAR}}{\overline{x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash x y y : \mathbb{B}} \text{APP}}}{\overline{x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B} \vdash \lambda y : \mathbb{B}. x y y : \mathbb{B} \rightarrow \mathbb{B}} \text{ABS}}}{\vdash \lambda x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}. \lambda y : \mathbb{B}. x y y : (\mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}} \text{ABS}$$

where Π_1 is

$$\frac{\overline{x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}} \text{VAR} \quad \overline{x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash y : \mathbb{B} \rightarrow \mathbb{B}} \text{VAR}}{\overline{x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash x y : \mathbb{B} \rightarrow \mathbb{B}} \text{APP}}$$

(d) $\lambda f : \mathbb{B} \rightarrow \mathbb{B}. \lambda x : \mathbb{B}. f(f(f(f(x))))$

$$\frac{\frac{\overline{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}} \text{VAR} \quad \frac{\frac{\frac{\frac{\Pi_1 \quad \overline{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f(f(f(f(x)))) : \mathbb{B}} \text{APP}}{\overline{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f(f(f(f(f(x)))) : \mathbb{B}} \text{APP}}}{\overline{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f(f(f(f(f(f(x)))) : \mathbb{B}} \text{ABS}}}{\overline{f : \mathbb{B} \rightarrow \mathbb{B} \vdash \lambda x : \mathbb{B}. f(f(f(f(f(x)))) : \mathbb{B}} \text{ABS}}}{\vdash \lambda f : \mathbb{B} \rightarrow \mathbb{B}. \lambda x : \mathbb{B}. f(f(f(f(f(x)))) : \mathbb{B}} \text{ABS}}$$

where Π_1 is

$$\frac{\frac{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{} \text{VAR} \quad \frac{\frac{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{} \text{VAR} \quad \Pi_2}{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f(f(x)) : \mathbb{B}} \text{APP}}{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f(f(f(x))) : \mathbb{B}} \text{APP}$$

where Π_2 is

$$\frac{\frac{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{} \text{VAR} \quad \frac{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash x : \mathbb{B}}{} \text{VAR}}{f : \mathbb{B} \rightarrow \mathbb{B}, x : \mathbb{B} \vdash f(x) : \mathbb{B}} \text{APP}$$

A better way to show that the term is typeable would be to prove the derivation exists by induction on the number of f 's.

(e) $\lambda x : \mathbb{B} \rightarrow \mathbb{B}. x \lambda y : \mathbb{B}. y$

Let us try to find a type derivation, using metavariables for types.

$$\frac{\frac{x : \mathbb{B} \rightarrow \mathbb{B} \vdash x : T_2}{} \text{VAR} \quad \frac{\frac{x : \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash y : T_4}{} \text{VAR} \quad \frac{x : \mathbb{B} \rightarrow \mathbb{B} \vdash \lambda y : \mathbb{B}. y : T_3}{} \text{ABS}}{\frac{x : \mathbb{B} \rightarrow \mathbb{B} \vdash x \lambda y : \mathbb{B}. y : T_1}{} \text{APP}} \text{ABS}$$

$$\frac{}{\vdash \lambda x : \mathbb{B} \rightarrow \mathbb{B}. x \lambda y : \mathbb{B}. y : T_0} \text{ABS}$$

This is the only possible shape for the typing derivation tree, since it follows the syntactic structure of term. But now we can deduce

$$T_2 = \mathbb{B} \rightarrow \mathbb{B}$$

$$T_4 = \mathbb{B}$$

since the VAR-rule has to make the type match the type from the typing environment. Inspecting the ABS-rule, we see that

$$T_3 = \mathbb{B} \rightarrow T_4 = \mathbb{B} \rightarrow \mathbb{B}$$

But now we are stuck, because for the APP-rule we need $T_2 = T_3 \rightarrow T_1$. But there is no T_1 satisfying $\mathbb{B} \rightarrow \mathbb{B} = (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow T_1$. Hence there is no typing derivation for the term.

(f) $\lambda x : \mathbb{B}. \lambda x : \mathbb{B} \rightarrow \mathbb{B}. x x$. We should α -rename to make the bound variables distinct:

$$\lambda x : \mathbb{B}. \lambda y : \mathbb{B} \rightarrow \mathbb{B}. y y.$$

Now, the only possible typing derivation would be

$$\frac{\frac{x : \mathbb{B}, y : \mathbb{B} \rightarrow \mathbb{B} \vdash y : T_3}{} \text{VAR} \quad \frac{x : \mathbb{B}, y : \mathbb{B} \rightarrow \mathbb{B} \vdash y : T_4}{} \text{VAR}}{\frac{x : \mathbb{B}, y : \mathbb{B} \rightarrow \mathbb{B} \vdash y y : T_2}{} \text{APP}} \text{VAR}$$

$$\frac{x : \mathbb{B} \vdash \lambda y : \mathbb{B} \rightarrow \mathbb{B}. y y : T_1}{} \text{ABS}$$

$$\frac{}{\vdash \lambda x : \mathbb{B}. \lambda y : \mathbb{B} \rightarrow \mathbb{B}. y y : T_0} \text{ABS}$$

for some types T_0, T_1, T_2, T_3, T_4 . From the VAR-rule instances at the top, $T_3 = \mathbb{B} \rightarrow \mathbb{B} = T_4$. But the APP-rule requires $T_3 = T_4 \rightarrow T_2$, i.e. that $\mathbb{B} \rightarrow \mathbb{B} = (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow T_2$, which is impossible.

(g) $\lambda x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}. \lambda y : \mathbb{B} \rightarrow \mathbb{B}. x \lambda z : \mathbb{B}. x y.$

We will write $\Gamma = x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}, y : \mathbb{B} \rightarrow \mathbb{B}$ to save space.

$$\frac{\frac{\frac{\frac{\frac{\Gamma, z : \mathbb{B} \vdash x : \mathbb{B} \rightarrow \mathbb{B}}{\Gamma \vdash x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{VAR}}{\Gamma \vdash x \lambda z : \mathbb{B}. x y : \mathbb{B}} \text{VAR}}{\Gamma \vdash \lambda z : \mathbb{B}. x y : \mathbb{B} \rightarrow \mathbb{B}} \text{ABS}}{\Gamma \vdash x \lambda z : \mathbb{B}. x y : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{ABS}} \text{APP} \text{ABS} \text{ABS}$$

(h) $\lambda z : \mathbb{B}. \text{if } z \text{ then } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. z \text{ else } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. f z$

$$\frac{\frac{\frac{\frac{\Gamma, z : \mathbb{B} \vdash z : \mathbb{B}}{\Gamma \vdash z : \mathbb{B}} \text{VAR}}{\Gamma \vdash \text{if } z \text{ then } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. z \text{ else } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. f z : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{ITE}}{\Gamma \vdash \lambda z : \mathbb{B}. \text{if } z \text{ then } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. z \text{ else } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. f z : \mathbb{B} \rightarrow (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{ABS}$$

where Π_1 is

$$\frac{\frac{\Gamma, z : \mathbb{B}, f : \mathbb{B} \rightarrow \mathbb{B} \vdash z : \mathbb{B}}{\Gamma \vdash \lambda f : \mathbb{B} \rightarrow \mathbb{B}. z : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{VAR} \text{ABS}$$

and Π_2 is

$$\frac{\frac{\frac{\Gamma, z : \mathbb{B}, f : \mathbb{B} \rightarrow \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{\Gamma \vdash \lambda f : \mathbb{B} \rightarrow \mathbb{B}. f z : \mathbb{B}} \text{VAR} \text{ABS}}{\Gamma \vdash \lambda f : \mathbb{B} \rightarrow \mathbb{B}. f z : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{APP}$$

Question 2: Typing open Church-style terms

For each of the following Church-style λ -terms t , find all typing contexts Γ and types T such that $\Gamma \vdash t : T$ is a valid typing judgement.

(a) x

For any type T , we can give x the type T in any context where $\Gamma(x) = T$:

$$\frac{}{\Gamma \vdash x : T} \text{VAR}$$

(b) xx

There are no pairs (Γ, T) for which $\Gamma \vdash xx : T$. A typing derivation would have to look like

$$\frac{\frac{\frac{\Gamma \vdash x : T_1 \rightarrow T}{\Gamma \vdash x : T_1} \text{VAR}}{\Gamma \vdash xx : T} \text{APP} \text{VAR}$$

for some T_1 . But we can't have both $\Gamma(x) = T_1$ and $\Gamma(x) = T_1 \rightarrow T$.

(c) $x y x$

This is not typable. Consider the form of a possible typing judgement.

$$\frac{\frac{\overline{\Gamma \vdash x : T_2 \rightarrow T_1 \rightarrow T} \text{ VAR} \quad \overline{\Gamma \vdash y : T_2} \text{ VAR}}{\Gamma \vdash x y : T_1 \rightarrow T} \text{ APP} \quad \overline{\Gamma \vdash x : T_1} \text{ VAR}}{\Gamma \vdash x y x : T} \text{ APP}$$

We need to have both $\Gamma(x) = T_1$ and $\Gamma(x) = T_2 \rightarrow T_1 \rightarrow T$, but this is impossible.

(d) $x (y x)$

We will see that this term can be given any type T in many different typing contexts. Consider the form of a possible typing judgement.

$$\frac{\overline{\Gamma \vdash x : T_1 \rightarrow T} \text{ VAR} \quad \frac{\overline{\Gamma \vdash y : T_2 \rightarrow T_1} \text{ VAR} \quad \overline{\Gamma \vdash x : T_2} \text{ VAR}}{\Gamma \vdash y x : T_1} \text{ APP}}{\Gamma \vdash x y x : T} \text{ APP}$$

These constraints actually admit solutions. We need

$$T_2 = T_1 \rightarrow T$$

and substituting this in we get

$$\frac{\overline{\Gamma \vdash x : T_1 \rightarrow T} \text{ VAR} \quad \frac{\overline{\Gamma \vdash y : (T_1 \rightarrow T) \rightarrow T_1} \text{ VAR} \quad \overline{\Gamma \vdash x : T_1 \rightarrow T} \text{ VAR}}{\Gamma \vdash y x : T_1} \text{ APP}}{\Gamma \vdash x y x : T} \text{ APP}$$

but this is actually a valid typing derivation for any types T and T_1 in a context Γ with $\Gamma(x) = T_1 \rightarrow T$ and $\Gamma(y) = (T_1 \rightarrow T) \rightarrow T_1$.

(e) $x \lambda y : \mathbb{B}. x y$.

Informally, this won't be typable because the outer x is applied to an abstraction term, whereas the inner x is applied to a bound variable of type $\mathbb{B}$. Consider

$$\frac{\overline{\Gamma \vdash x : T_1 \rightarrow T} \text{ VAR} \quad \frac{\frac{\overline{\Gamma, y : \mathbb{B} \vdash x : T_3 \rightarrow T_2} \text{ VAR} \quad \overline{\Gamma, y : \mathbb{B} \vdash y : T_3} \text{ VAR}}{\Gamma, y : \mathbb{B} \vdash x y : T_2} \text{ APP}}{\Gamma \vdash \lambda y : \mathbb{B}. x y : T_1} \text{ ABS}}{\Gamma \vdash x \lambda y : \mathbb{B}. x y : T} \text{ APP}$$

where we must have $T_1 = \mathbb{B} \rightarrow T_2$ for the ABS-instance to be valid. We can also easily see that $T_3 = \mathbb{B}$, for one of the VAR-instances to be valid. Let's substitute those in

$$\frac{\overline{\Gamma \vdash x : (\mathbb{B} \rightarrow T_2) \rightarrow T} \text{ VAR} \quad \frac{\frac{\overline{\Gamma, y : \mathbb{B} \vdash x : \mathbb{B} \rightarrow T_2} \text{ VAR} \quad \overline{\Gamma, y : \mathbb{B} \vdash y : \mathbb{B}} \text{ VAR}}{\Gamma, y : \mathbb{B} \vdash x y : T_2} \text{ APP}}{\Gamma \vdash \lambda y : \mathbb{B}. x y : \mathbb{B} \rightarrow T_2} \text{ ABS}}{\Gamma \vdash x \lambda y : \mathbb{B}. x y : T} \text{ APP}$$

Now we need the types of x in the two VAR rules to agree, $\mathbb{B} \rightarrow T_2 = (\mathbb{B} \rightarrow T_2) \rightarrow T$, but this is impossible.

(f) $\lambda y : \mathbb{B}. x (x y)$.

Informally, we see x is applied to a bound variable known to be of type $\mathbb{B}$. Then x is applied to result of that, so the result of x should also be of type $\mathbb{B}$. So the only possible type should be $\mathbb{B} \rightarrow \mathbb{B}$, in a context where $x : \mathbb{B} \rightarrow \mathbb{B}$.

More formally, consider the following derivation. Since the term is a λ -abstraction, we'll immediately swap out T for $\mathbb{B} \rightarrow T_1$.

$$\frac{\frac{\frac{\Gamma, y : \mathbb{B} \vdash x : T_2 \rightarrow T_1}{\Gamma, y : \mathbb{B} \vdash x : T_2 \rightarrow T_1} \text{VAR} \quad \frac{\frac{\Gamma, y : \mathbb{B} \vdash x : T_3 \rightarrow T_2}{\Gamma, y : \mathbb{B} \vdash x : T_3 \rightarrow T_2} \text{VAR} \quad \frac{\Gamma, y : \mathbb{B} \vdash y : T_3}{\Gamma, y : \mathbb{B} \vdash y : T_3} \text{VAR}}{\Gamma, y : \mathbb{B} \vdash x y : T_2} \text{APP} \quad \frac{\Gamma, y : \mathbb{B} \vdash x (x y) : T_1}{\Gamma \vdash \lambda y : \mathbb{B}. x (x y) : \mathbb{B} \rightarrow T_1} \text{ABS}$$

Having got to the top of the tree, we see that $T_3 = \mathbb{B}$. We need to have

$$\mathbb{B} \rightarrow T_2 = T_2 \rightarrow T_1.$$

This is the case iff $T_2 = \mathbb{B}$ and $T_1 = T_2 = \mathbb{B}$. This proves that $\mathbb{B} \rightarrow \mathbb{B}$ is the only possible type the term could have.

But with those values, we do have a valid typing derivation:

$$\frac{\frac{\frac{\Gamma, y : \mathbb{B} \vdash x : \mathbb{B} \rightarrow \mathbb{B}}{\Gamma, y : \mathbb{B} \vdash x : \mathbb{B} \rightarrow \mathbb{B}} \text{VAR} \quad \frac{\frac{\Gamma, y : \mathbb{B} \vdash x : \mathbb{B} \rightarrow \mathbb{B}}{\Gamma, y : \mathbb{B} \vdash x : \mathbb{B} \rightarrow \mathbb{B}} \text{APP} \quad \frac{\Gamma, y : \mathbb{B} \vdash y : \mathbb{B}}{\Gamma, y : \mathbb{B} \vdash y : \mathbb{B}} \text{APP}}{\Gamma, y : \mathbb{B} \vdash x y : \mathbb{B}} \text{APP} \quad \frac{\Gamma, y : \mathbb{B} \vdash x (x y) : \mathbb{B}}{\Gamma \vdash \lambda y : \mathbb{B}. x (x y) : \mathbb{B} \rightarrow \mathbb{B}} \text{ABS}$$

for any context where $\Gamma(x) = \mathbb{B} \rightarrow \mathbb{B}$.

(g) **if** x **then** x **else** x .

Informally, x needs to be of type $\mathbb{B}$ because it appears in the guard of the 'if' statement. Therefore the whole term will have to be of type $\mathbb{B}$.

Formally, the only possible typing derivation looks like

$$\frac{\frac{\Gamma \vdash x : \mathbb{B}}{\Gamma \vdash x : \mathbb{B}} \text{VAR} \quad \frac{\Gamma \vdash x : T}{\Gamma \vdash x : T} \text{VAR} \quad \frac{\Gamma \vdash x : T}{\Gamma \vdash x : T} \text{VAR}}{\Gamma \vdash \text{if } x \text{ then } x \text{ else } x : T} \text{ITE}$$

and this is a valid derivation iff $T = \mathbb{B}$ and $\Gamma(x) = \mathbb{B}$.

(h) $z (y \text{ true}) (y x) (x \text{ false})$

Informally, this won't be typable. The z only appears once and at the head of the term, so it will clearly be a three-argument function with a free choice of codomain and domains forced by the other subterms. Since different instances of y are applied to **true** and x , x must have type $\mathbb{B}$ and y must have a type $\mathbb{B} \rightarrow T_1$ for some T_1 . But there is also a subterm $x \text{ false}$, which is not possible if $x : \mathbb{B}$.

Formally, consider

$$\frac{\frac{\frac{\Gamma \vdash x : \mathbb{B} \rightarrow T_1}{\Gamma \vdash x : \mathbb{B} \rightarrow T_1} \text{VAR} \quad \frac{\Gamma \vdash \text{false} : \mathbb{B}}{\Gamma \vdash \text{false} : \mathbb{B}} \text{F}}{\Gamma \vdash x \text{ false} : T_1} \text{APP} \quad \frac{\Pi_1}{\Gamma \vdash z (y \text{ true}) (y x) (x \text{ false}) : T} \text{APP}$$

where Π_1 is

$$\frac{\frac{\frac{}{\Gamma \vdash y : (\mathbb{B} \rightarrow T_1) \rightarrow T_2} \text{VAR}}{\Pi_2 \quad \Gamma \vdash y x : T_2} \text{APP}}{\Gamma \vdash z (y \text{true}) (y x) : T_1 \rightarrow T} \text{APP}$$

where here we have used the constraint on $\Gamma(x)$ deduced from the other part of the derivation, and where Π_2 is

$$\frac{\frac{\frac{}{\Gamma \vdash z : T_3 \rightarrow T_1 \rightarrow T} \text{VAR}}{\Gamma \vdash z (y \text{true})} \text{APP}}{\Gamma \vdash z (y \text{true})} \text{APP}$$

Now we need both $\Gamma(y) = \mathbb{B} \rightarrow T_3$ and $\Gamma(y) = (\mathbb{B} \rightarrow T_1) \rightarrow T_2$, which requires $\mathbb{B} = \mathbb{B} \rightarrow T_1$, which is impossible.